

EXP NO: 9	Basic Calculator App Using Android UI Controls
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AIM:

Develop an Android application using controls like Button, TextView, EditText for designing a calculator having basic functionality like Addition, Subtraction, multiplication, and Division.

CODE:

MainActivity.kt

```
package com.example.myapplication_karthick_95

import android.os.Bundle
import android.view.View
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    private lateinit var display: EditText
    private var currentInput: String = ""
    private var operand1: Double = 0.0
    private var operand2: Double = 0.0
    private var operator: String = ""

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        display = findViewById(R.id.display)

        // Number buttons
        setButtonClickListener(R.id.button0)
        setButtonClickListener(R.id.button1)
        setButtonClickListener(R.id.button2)
        setButtonClickListener(R.id.button3)
        setButtonClickListener(R.id.button4)
        setButtonClickListener(R.id.button5)
        setButtonClickListener(R.id.button6)
        setButtonClickListener(R.id.button7)
        setButtonClickListener(R.id.button8)
        setButtonClickListener(R.id.button9)

        // Operator buttons
        setOperatorClickListener(R.id.buttonAdd, "+")
        setOperatorClickListener(R.id.buttonSubtract, "-")
        setOperatorClickListener(R.id.buttonMultiply, "*")
        setOperatorClickListener(R.id.buttonDivide, "/")
    }
}
```

```

// Clear button
findViewById<Button>(R.id.buttonClear).setOnClickListener {
    currentInput = ""
    operand1 = 0.0
    operand2 = 0.0
    operator = ""
    display.setText("")
}

// Equal button
findViewById<Button>(R.id.buttonEqual).setOnClickListener {
    if (operator.isEmpty()) return@setOnClickListener

    try {
        operand2 = currentInput.toDouble()

        val result = when (operator) {
            "+" -> operand1 + operand2
            "-" -> operand1 - operand2
            "*" -> operand1 * operand2
            "/" -> {
                if (operand2 == 0.0) {
                    Toast.makeText(this@MainActivity, "Cannot divide by zero",
Toast.LENGTH_SHORT).show()
                    return@setOnClickListener
                }
                operand1 / operand2
            }
            else -> 0.0
        }

        display.setText(result.toString())
        operand1 = result // Update operand1 for subsequent calculations
        operator = ""
        currentInput = result.toString()
    } catch (e: Exception) {
        Toast.makeText(this@MainActivity, "Error: Invalid Input",
Toast.LENGTH_SHORT).show()
    }
}

// Set up number button listeners
private fun setButtonClickListener(buttonId: Int) {
    val button: Button = findViewById(buttonId)
    button.setOnClickListener {
        currentInput += button.text.toString()
        display.setText(currentInput)
    }
}

// Set up operator button listeners
private fun setOperatorClickListener(buttonId: Int, op: String) {
    val button: Button = findViewById(buttonId)

```

```

        button.setOnClickListener {
            if (currentInput.isNotEmpty()) {
                operand1 = currentInput.toDouble()
                currentInput = ""
                operator = op
            }
        }
    }

    // Optional: Add functionality to handle decimal point (if needed)
    private fun isDecimalPointValid(): Boolean {
        return !currentInput.contains(".")
    }
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/calculatorLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/display"
        android:layout_width="match_parent"
        android:layout_height="80dp"
        android:background="#000000"
        android:textColor="#FFFFFF"
        android:textSize="28sp"
        android:gravity="end|center_vertical"
        android:inputType="none"
        android:focusable="false"
        android:clickable="false"
        android:padding="12dp"
        android:layout_marginBottom="12dp" />

    <!-- Row 1: 7 8 9 / -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button android:id="@+id/button7" style="@style/CalcButton" android:text="7" />
        <Button android:id="@+id/button8" style="@style/CalcButton" android:text="8" />
        <Button android:id="@+id/button9" style="@style/CalcButton" android:text="9" />
        <Button android:id="@+id/buttonDivide" style="@style/CalcButton" android:text="/" />
    </LinearLayout>

    <!-- Row 2: 4 5 6 * -->
    <LinearLayout

```

```

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        " android:orientation="horizontal">
        <Button android:id="@+id/button4" style="@style/CalcButton" android:text="4" />
        <Button android:id="@+id/button5" style="@style/CalcButton" android:text="5" />
        <Button android:id="@+id/button6" style="@style/CalcButton" android:text="6" />
        <Button android:id="@+id/buttonMultiply" style="@style/CalcButton" android:text="*" />
    </LinearLayout>

    <!-- Row 3: 1 2 3 - -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button android:id="@+id/button1" style="@style/CalcButton" android:text="1" />
        <Button android:id="@+id/button2" style="@style/CalcButton" android:text="2" />
        <Button android:id="@+id/button3" style="@style/CalcButton" android:text="3" />
        <Button android:id="@+id/buttonSubtract" style="@style/CalcButton" android:text="-" />
    </LinearLayout>

    <!-- Row 4: 0 C = + -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button android:id="@+id/button0" style="@style/CalcButton" android:text="0" />
        <Button android:id="@+id/buttonClear" style="@style/CalcButton" android:text="C" />
        <Button android:id="@+id/buttonEqual" style="@style/CalcButton" android:text="=" />
        <Button android:id="@+id/buttonAdd" style="@style/CalcButton" android:text="+" />
    </LinearLayout>

</LinearLayout>

```

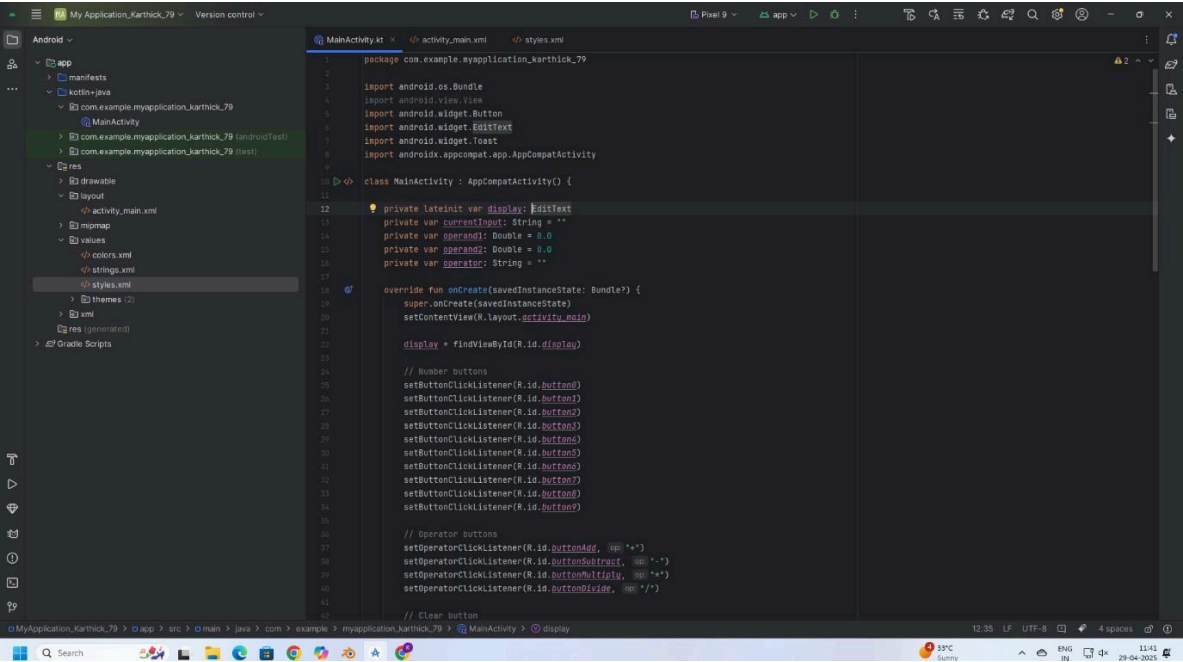
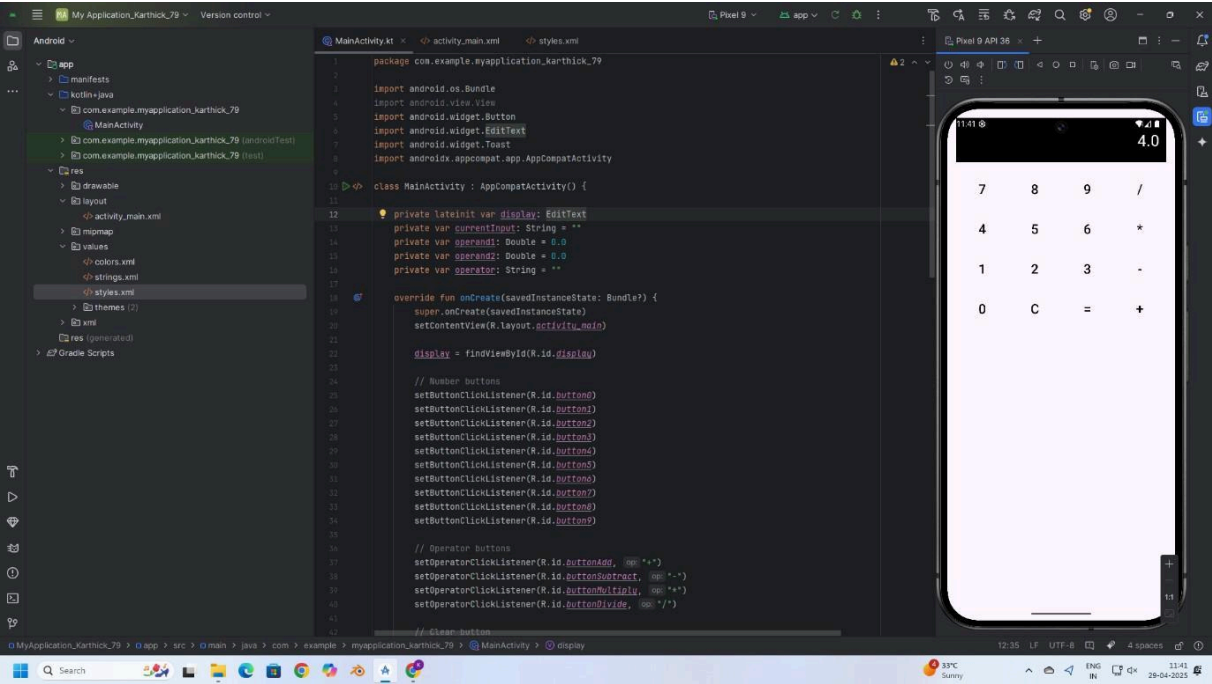
styles.xml

```

<resources>
<style name="CalcButton">
<item name="android:layout_width">0dp</item>
<item name="android:layout_height">wrap_content</item>
<item name="android:layout_weight">1</item>
<item name="android:padding">16dp</item>
<item name="android:textSize">24sp</item>
<item name="android:background">?attr/selectableItemBackground</item>
<item name="android:textColor">#000000</item>
</style>
</resources>

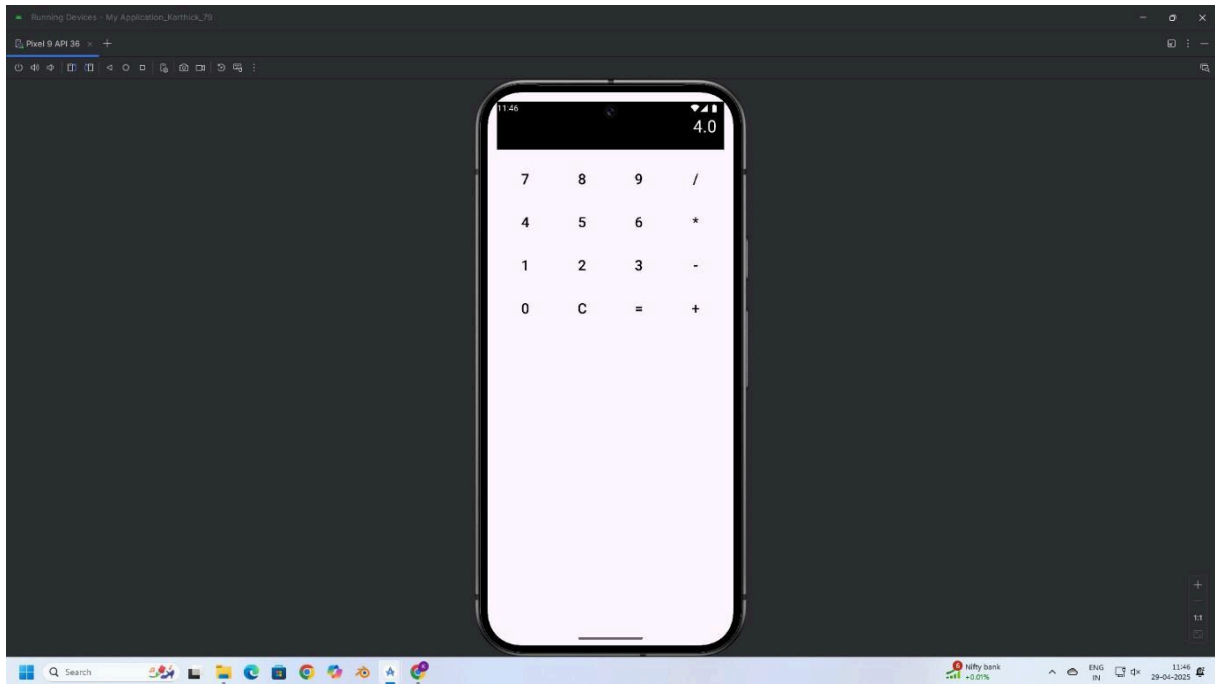
```

OUTPUT:



```
1 package com.example.myapplication_karthick_79
2
3 import androidx.appcompat.app.AppCompatActivity
4 import android.os.Bundle
5 import android.view.View
6 import android.widget.Button
7 import android.widget.EditText
8
9 class MainActivity : AppCompatActivity() {
10     override fun onCreate(savedInstanceState: Bundle?) {
11         setContentView(R.layout.activity_main)
12     }
13
14     fun calculate() {
15         // TODO: Implement the calculation logic
16     }
17
18     fun add() {
19         // TODO: Implement the addition logic
20     }
21
22     fun subtract() {
23         // TODO: Implement the subtraction logic
24     }
25
26     fun multiply() {
27         // TODO: Implement the multiplication logic
28     }
29
30     fun divide() {
31         // TODO: Implement the division logic
32     }
33
34     fun clear() {
35         // TODO: Implement the clear logic
36     }
37
38     fun equals() {
39         // TODO: Implement the equals logic
40     }
41
42     fun digit(digit: Int) {
43         // TODO: Implement the digit input logic
44     }
45
46     fun decimal() {
47         // TODO: Implement the decimal input logic
48     }
49
50     fun percent() {
51         // TODO: Implement the percent input logic
52     }
53
54     fun inverse() {
55         // TODO: Implement the inverse logic
56     }
57
58     fun power() {
59         // TODO: Implement the power logic
60     }
61
62     fun squareRoot() {
63         // TODO: Implement the square root logic
64     }
65
66     fun log() {
67         // TODO: Implement the log logic
68     }
69
70     fun log10() {
71         // TODO: Implement the log10 logic
72     }
73
74     fun sin() {
75         // TODO: Implement the sin logic
76     }
77
78     fun cos() {
79         // TODO: Implement the cos logic
80     }
81
82     fun tan() {
83         // TODO: Implement the tan logic
84     }
85
86     fun cot() {
87         // TODO: Implement the cot logic
88     }
89
90     fun sec() {
91         // TODO: Implement the sec logic
92     }
93
94     fun csc() {
95         // TODO: Implement the csc logic
96     }
97
98     fun sinh() {
99         // TODO: Implement the sinh logic
100     }
101
102     fun cosh() {
103         // TODO: Implement the cosh logic
104     }
105
106     fun tanh() {
107         // TODO: Implement the tanh logic
108     }
109
110     fun coth() {
111         // TODO: Implement the coth logic
112     }
113
114     fun sech() {
115         // TODO: Implement the sech logic
116     }
117
118     fun csch() {
119         // TODO: Implement the csch logic
120     }
121
122     fun erf() {
123         // TODO: Implement the erf logic
124     }
125
126     fun erfc() {
127         // TODO: Implement the erfc logic
128     }
129
130     fun gamma() {
131         // TODO: Implement the gamma logic
132     }
133
134     fun digamma() {
135         // TODO: Implement the digamma logic
136     }
137
138     fun zeta() {
139         // TODO: Implement the zeta logic
140     }
141
142     fun beta() {
143         // TODO: Implement the beta logic
144     }
145
146     fun psi() {
147         // TODO: Implement the psi logic
148     }
149
150     fun polygamma() {
151         // TODO: Implement the polygamma logic
152     }
153
154     fun erfci() {
155         // TODO: Implement the erfci logic
156     }
157
158     fun erfcx() {
159         // TODO: Implement the erfcx logic
160     }
150 }
```

```
1 <resources>
2     <style name="CalcButton">
3         <item name="android:layout_width">wrap_content</item>
4         <item name="android:layout_height">wrap_content</item>
5         <item name="android:padding">10dp</item>
6         <item name="android:textSize">24sp</item>
7         <item name="android:background">@drawable/selectableItemBackground</item>
8         <item name="android:textColor">@android:color/black</item>
9     </style>
10 </resources>
```



RESULT:

Thus, a basic calculator application was successfully developed using Android controls like Button, TextView, and EditText to perform addition, subtraction, multiplication, and division operations.